13 CONSTRUCTORS AND DESTRUCTORS • THEORY

01 constructors types and initializer lists

Detailed Theory, Exact C++ Rules, Memory Models & Exam Reference

01 — Constructors, Constructor Overloading, and Member Initializer Lists

1. Properties of a Constructor

- Has the **exact same name** as the class.
- Has **no return type** (not even `void`).
- Invoked automatically when an object is created.
- If you define **any** parameterized constructor, the compiler **suppresses** the implicitly-declared default constructor (unless you restore it with `person() = default;`).

2. When Member Initializer Lists (`: member(val)`) Are Mandatory

1. Initializing `const` **data members** (`const int id;`).
2. Initializing **reference data members** (`int& ref;`).
3. Initializing **base classes** or **member objects** that lack a default constructor.
4. **Initialization Order Rule**: Data members are always initialized in the **exact order they are declared inside the class**, **not** the order written in the member initializer list!